

U.G.C. NET Exam

COMPUTER

(Solved with Explanation)

1. If the primal linear programming problem has unbounded solution, then it's dual problem will have :
- feasible solution
 - alternative solution
 - no feasible solution at all
 - no bounded solution at all

Ans : (c) If the primal problem is feasible, but unbounded in the direction of optimization, then the dual has no feasible solution otherwise, if the primal problem has an optimal solution, then dual will also have an optimal solution.

2. Given the problem to maximize

$$f(x), X = (x_1, x_2, \dots, x_n)$$

subject to m number of inequality constraints.

$$g_i(x) \leq b_i, i = 1, 2, \dots, m$$

including the non-negativity constraints $x \geq 0$.

Which of the following conditions is a Kuhn-Tucker necessary condition for a local maxima at $\bar{x}$?

- $\frac{\partial L(\bar{X}, \bar{\lambda}, \bar{S})}{\partial x_j} = 0, j = 1, 2, \dots, m$
- $\bar{\lambda}_i [g_i(\bar{X}) - b_i] = 0, i = 1, 2, \dots, m$
- $g_i(\bar{X}) \leq b_i, i = 1, 2, \dots, m$
- All of the above

Ans : (d) Kuhn-Tucker necessary conditions:

The necessary conditions for a local maxima or stationary point(s) at $\bar{x}$ are.

- $\frac{\delta L}{\delta x_j}(\bar{x}, \bar{\lambda}, \bar{\delta}) = 0, j = 1, 2, \dots, n$
- $\bar{\lambda}_i [g_i(\bar{x}) - b_i] = 0$
- $g_i(\bar{x}) \leq b_i$
- $\bar{\lambda}_i \geq 0, i = 1, 2, \dots, m$

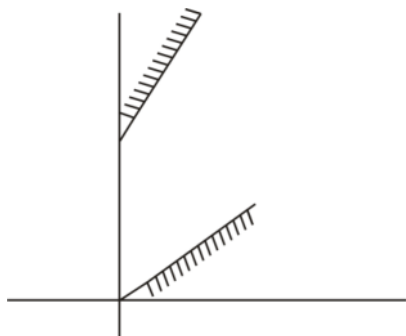
3. The following linear programming problem has $\text{Max } Z = x_1 + x_2$
 Subject to $x_1 - x_2 \geq 0$
 $3x_1 - x_2 \leq -3$
 and $x_1, x_2 \geq 0$
- Feasible solution
 - No feasible solution
 - Unbounded solution
 - single point as solution

n. December, 2013

R SCIENCE

anation Paper–III)

Ans : (b) draw lines for two inequalities nothing is common in shaded area so infeasible solution.



$$3x_1 - x_2 + 3 \leq 0$$

$$x_1 - x_2 \geq 0$$

4. Given a flow graph with 10 nodes, 13 edges and one connected components, the number of regions and the number of predicate (decision) nodes in the flow graph will be

- (a) 4, 5 (b) 5, 4
(c) 3, 1 (d) 13, 8

Ans : (b) 5, 4

Predicate (decision) nodes are those nodes which has out degree at least 2.

Region (2) $13 - 10 + 12 = 5$ [i.e. edges – vertex + (P + 1)] since P = 1

Predicate nodes (P) = region = P + 1 = 5

P = 4

5. Function points can be calculated by :

- (a) UFP * CAF
(b) UFP * FAC
(c) UFP * Cost
(d) UFP * Productivity

Ans : (a) UFP * CAF

Function points = UFP * CAF

where UFP = unadjusted function points

CAF = complexity adjustment factor

6. Match the following :

List-I		List-II	
A.	Data coupling	1.	Module A and Module B have shared data
B.	Stamp coupling	2.	Dependency between modules is based on the fact they communicate by only passing of data

C.	Common coupling	3.	When complete data structure is passed from one module to another
D.	Content coupling	4.	When the control is passed from one module to the middle of another

	A	B	C	D
(a)	3	2	1	4
(b)	2	3	1	4
(c)	2	3	4	1
(d)	3	2	4	1

Ans : (b)

(a) Data coupling is the depends between modules is based on the fact they can communicate by only passing of data, so it satisfies (2)

(b) Stamp coupling is when compute data structure is passed from one module to another is the stamp coupling so it satisfies (3)

(c) Common coupling is a two modules that have shared a data, so it satisfies (1)

(d) Content coupling is when the control is passed from one module to the middle of another so it satisfies (4)

7. A process which defines a series of tasks that have the following four primary objectives is known as

- (1) to identity all items that collectively define the software configuration**
 - (2) to manage change to one or more of these items**
 - (3) to facilitate the construction of different versions of an application.**
 - (4) to ensure that software quality is maintained as the configuration evolves over time.**
- (a) Software quality management process
 - (b) Software configuration management process
 - (c) Software version management process
 - (d) Software change management process

Ans : (b) Software configuration management process has following primary objective :

1. to identify all item that collectively define the software configuration
2. to manage changes to one or more of those item
3. to facilitate the construction of different version of an application
4. to ensure that software quality is maintained as the configuration evolves over time.

8. One weakness of boundary value analysis and equivalence partitioning is :

- (a) they are not effective
- (b) they do not explore combinations of input circumstances
- (c) they explore combinations of input circumstances
- (d) none of the above

Ans : (b) Both are black box testing techniques as we only test at boundaries and at certain points.
Dividing the test input data into a range of values and selecting one input from each range is called Equivalence partition. It is not explore combination of input circumstances.

- 9. Which once of the following is not a software myth?**
- (a) Once we write the program and get it to work, our job is done
 - (b) Project requirements continually change, but change can be easily accommodated because software is flexible
 - (c) If we get behind schedule, we can add more programmers and catch up
 - (d) If an organization does not understand how to control software projects internally, it will invariably struggle when it outsources software projects

Ans : (d) Software myth is all people who come in contact with software may suffer from various myths associated with developing and using software, Management myths, customer myths, programmer myths, are the few common myths,
Here in question (a), (b), (c) are myths (programmer & managements myths).

- 10. Match the following with respect to relationship between objects and classes :**

List-I		List-II	
A.	State diagram	1.	Useful for both abstract modeling and for designing actual program
B.	Object diagram	2.	Describes object classes
C.	Class diagram	3.	Useful for documenting test cases
D.	Instance diagram	4.	Describing the behavior of a single class of objects

	A	B	C	D
(a)	4	1	2	3
(b)	2	3	4	1
(c)	3	4	2	1
(d)	2	4	1	3

Ans : (a)
(a) State diagram is a type of diagram used in computer science and related fields to describe the behavior of systems. State diagrams require that the system described is composed of a finite number of states. So it satisfies (4)
(b) Object diagram are derived from class diagram so object diagram are dependent upon class diagrams
object diagram represent an instance of a class diagram

also represent the static view of a system but this static view is a snapshot of the system at a particular moment.
(c) Class diagram is united modeling language (UML) is a type of static structure diagram that describes the structure of a system. So it satisfies (2)
(d) Instance diagram is a part of the unified modeling language that one does not mentioned too after the basic idea is to make a static snap shot of instance in your system.

11. Match the following style rules for reusability :

List-I		List-II	
A.	Keep methods coherent	1.	Write a method to get the last element of a list
B.	Keep methods small	2.	Maintain parallel structure when possible
C.	Keep methods consistent	3.	Breaking a method into smaller parts
D.	Provide uniform coverage	4.	Performs a single function or a group of closely related functions

	A	B	C	D
(a)	4	2	3	1
(b)	2	1	4	3
(c)	3	4	2	1
(d)	2	3	4	1

Ans : (a)

- (a) Keep methods coherent to perform a single function or a group of a closely related functions, so it satisfies (4)
(b) Keep method small is breaking a method into smaller parts so it satisfies (3)
(c) Keep methods consistent to maintain parallel structure when possible. So it satisfies (2)
(d) Provide uniform coverage to write a method to get the last element of a list, so it satisfies (1)

12. Which is the protocol for performing RPCs between applications in a language and system independent way?

- (a) Hyper Text Transmission Protocol (HTTP)
(b) Simple Network Management Protocol (SNMP)
(c) Simple Object Access Protocol (SOAP)
(d) Simple Mail Transfer Protocol (SMTP)

Ans : (c) SOAP is an acronym for simple object access protocol. It is an XML based messaging protocol for exchanging information among computers. SOAP is an application of the XML Specification.

Although SOAP can be used in a variety of messaging system & can be delivered via a transport protocols, the

initial focus of SOAP is remote procedure calls (RPC) transported via HTTP.

- 13. The document that is used by SSLT to indicate, how to transform the elements of the XML document to another format is:**
- (a) HTML page
 - (b) DOC type procedure
 - (c) Style sheet
 - (d) Stored procedure

Ans : (c) Under XSLT, A XLST stylesheet is used to describe transformation rules in XML format. This is read by an application called an "XSLT Processor", transforming a designated XML document. The transformation results are output in XML, HTML or text format.

- 14. Which of the following concepts means adding new concepts to a program as it runs ?**
- (a) Data hiding
 - (b) Dynamic loading
 - (c) Dynamic typing
 - (d) Dynamic binding

Ans : (b) Dynamic loading is a mechanism by which a computer program can, at run time, load a library into memory. Retrieve the addresses of functions & variables contained in the library, execute those function or access those values & unload the library from memory.

In oops dynamic binding reasons to linking a procedure call to the code that will executed only at run time. The code associated with the procedure is not known until the program is executed, which is also known as late binding.

- 15. Which of the following correctly describes overloading of functions?**
- (a) Virtual polymorphism
 - (b) Transient polymorphism
 - (c) Ad-hoc polymorphism
 - (d) Pseudo polymorphism

Ans : (c) Ad – hoc polymorphism allows function with same name act differently for each type

In programming language, ad hoc polymorphism is a kind of polymorphism in which functions can be applied to arguments of different types, because a polymorphic function can denote a number of distinct & potentially heterogeneous implementation depending on the type of arguments (s) to which it is applied.

- 16. Match the following with respect to programming languages :**

List-I		List-II	
A.	Structured Language	1.	JAVA
B.	Non-Structured Language	2.	BASIC
C.	Object Oriented Programming Language	3.	PASCAL

D.	Interpreted Programming Language	4.	FORTRAN
----	----------------------------------	----	---------

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	2	1
(c)	2	4	1	3
(d)	2	3	4	1

Ans : (a) **PASCAL** - is a structured language
FORTRAN - is a non-structured language
JAVA - is an object oriented language
BASIC - is interpreted programming language.

17. The compiler converts all operands upto the type of the largest operand is called

- (a) Type Promotion (b) Type Evaluation
(c) Type Conversion (d) Type Declaration

Ans : (a) When constant and variable of different types are mixed in an expression, they are all converted to the same type. The compiler converts all operands up to the type of the largest operand, which is called type promotion.

18. C⁺⁺ actually supports the following two complete dynamic systems :

- (a) One defined by C⁺⁺ and the other not defined by C
(b) One defined by C and one specific to C⁺⁺
(c) Both are specific to C⁺⁺
(d) Both of them are improvements of C

Ans : (b) C⁺⁺ actually supports two complete dynamic allocation system: the one defined by C and the one specific to C⁺⁺. The system specific to C⁺⁺ contain several improvements over that.

19. Important advantage of using new and delete operators in C⁺⁺ is :

- (a) Allocation of memory
(b) Frees the memory previously allocated
(c) Initialization of memory easily
(d) Allocation of memory and frees the memory previously allocated

Ans : (d) C⁺⁺ supports dynamic allocation and deallocation of objects using the new and delete operators. These operators allocate memory for objects from a pool called the free-store. The new operator calls the special function operator 'new' and the delete operator calls the special function operator! "delete".

20. Match the following control strategies of prolog–

List-I		List-II	
A.	Forward movement	1.	Variable can be done with a constant, another variable or a function

B.	Unification	2.	The entire conjunctive goal is executed
C.	Deep backtracking	3.	Previous sub goal to find alternative solutions
D.	Shallow backtracking	4.	Chooses sub goal with possible unifier

	A	B	C	D
(a)	4	1	2	3
(b)	2	4	1	3
(c)	3	1	4	2
(d)	2	3	4	1

Ans : (a)

- (a) Forward movement is choose a rule by remember the position of the matched rule, so it satisfies (4)
 (b) Unification is a process of matching or finding the most general unifier. So it satisfies (1)
 (c) Deep back tracking is a previous sub goal to find alternative solutions, so it satisfies (2)
 (d) Shadow backtracking is the entire conjunctive goal's execution (3)

21. Given the following statements:

S1 : The grammars $S \rightarrow aSb|bSa|SS|a$ and $S \rightarrow aSb|bSa|a$ are not equivalent.

S2: The grammars $S \rightarrow SS|SSS|aSb|bSa|\lambda$ and $S \rightarrow SS|aSb|bSa|\lambda$ are equivalent.

Which of the following is true?

- (a) S1 is correct and S2 is not correct
 (b) Both S1 and S2 are correct
 (c) S1 is not correct and S2 is correct
 (d) Both S1 and S2 are not correct

Ans : (b) S1: Given grammars are

$$G_1 : S \rightarrow aSb|bSa|SS|a$$

and $G_2 : S \rightarrow aSb|bSa|a$

Grammar G_1 can generate string $w = aa$, but

Grammar G_2 can not generate this string.

So, G_1 and G_2 are not equivalent.

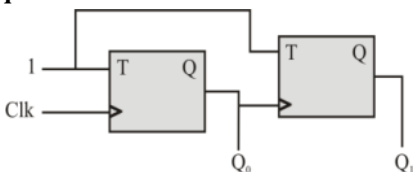
S2: Given grammar are

$$G_3 : S \rightarrow SS|SSS|aSb|bSa|\lambda$$

and $G_4 : S \rightarrow SS|aSb|bSa|\lambda$

Both grammar G_3 and G_4 are generating set of all string with equal number of a's and b's.

22. What are the final values of Q_1 and Q_0 after 4 clock cycles, if initial values are 00 in the sequential circuit shown below:



- (a) 11 (b) 10
(c) 01 (d) 00

Ans : (d) For T -flip-flop

T	Q^+
0	Q
1	$\bar{Q}$

Initial: 11 state Q_0 changed to 1 because T is 1 and previous state was 0 and state Q_1 changed to 1 because input T is 1 and previous state was 0.

Similarly,

Clock 2: 01

Clock 3: 10

Clock 4: 00

- 23. High level knowledge which relates to the use of sentences in different context and how the context affect the meaning of the sentences?**

- (a) Morphological (b) Syntactic
(c) Semantic (d) Pragmatic

Ans : (d) **Syntactic:** relating to the structure of statement or elements in a computer language.

Semantic: of words and language connected with meaning.

Progamatics: is the third general area of language description, refering to practical aspects of how constructs and features of language may be used to achieve various objectives.

- 24. The objective of procedure is to discover at least one that cause two literals to match.**

- (a) unification, validation
(b) unification, substitution
(c) Substitution, unification
(d) minimum, maximum

Ans : (b) In propositional logic unification also need one or more substitution that cause two literals like P & $\sim p$ to match so that they cancel out each other.

- 25. If h^* represent an estimate of the cost of getting from the current node N to the goal node and h represents actual cost of getting from the current node the goal node, then A^* algorithm gives an optimal solution if**

- (a) h^* is equal to h
(b) h^* overestimates h
(c) h^* underestimates h
(d) None of these

Ans : (c) Rule is algorithm gives an optimal solution if h^* never over estimates h (admissible heuristics) so b is out.

Underestimating h means algo is looking better solution which may be found if these are optimal solution.

26. The mean-end analysis process centers around the detection of differences between the current state and goal state. Once such a difference is isolated, an operator that can reduce the difference must be found. But perhaps that operator cannot be applied to the current state. So a sub-problem of getting to a state in which it can be applied is set up. The kind of backward chaining in which operators are selected and then sub goals are set up to establish the precondition of operators is called
- (a) backward planning
 - (b) goal stack planning
 - (c) operator subgoalings
 - (d) operator overloading

Ans : (c) A computer problem-solving method in which the inability of the computer to take the desired next step at any point in the problem solving process leads to a subgoal of making that step feasible.

27. In alpha-beta pruning, is used to cut off the search at maximizing level only and is used to cut off the search at minimizing level only.
- (a) alpha, beta
 - (b) beta, alpha
 - (c) alpha, alpha
 - (d) beta, beta

Ans : (b) Alpha beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the minimum algorithm in its search tree. Let's define the parameters Alpha and beta-
Alpha- is the best value that the maximizer currently can guarantee at that level or above.
Beta- is the best value that the minimizer currently can guarantee at the level or above. So answer is (b)

28. If A and B are two fuzzy sets with membership functions
- $\mu_A(x) = \{0.2, 0.5, 0.6, 0.1, 0.9\}$
 $\mu_B(x) = \{0.2, 0.5, 0.6, 0.7, 0.8\}$
 Then the value of $\mu_{A \cap B}$ will be
- (a) {0.2, 0.5, 0.6, 0.7, 0.9}
 - (b) {0.2, 0.5, 0.2, 0.1, 0.8}
 - (c) {0.1, 0.5, 0.6, 0.1, 0.8}
 - (d) {0.1, 0.5, 0.2, 0.1, 0.8}

Ans : (d) Intersection of two fuzzy sets-
 $\mu A(x) \cap \mu B(x) = \mu A(x) \wedge \mu B(x) = \min(\mu A(x), \mu B(x))$
 $\mu A(x) = \{0.2, 0.5, 0.6, 0.1, 0.9\}$
 $\mu B(x) = \{0.1, 0.5, 0.2, 0.7, 0.8\}$
 $\mu A \cap B(x) = \{0.1, 0.5, 0.2, 0.1, 0.8\}$

29. The height $h(A)$ of a fuzzy set A is defined as
- $h(A) = \sup A(x)$
 $x \in A$
 Then the fuzzy set A is called normal when
- (a) $h(A) = 0$
 - (b) $h(A) < 0$
 - (c) $h(A) = 1$
 - (d) $h(A) < 1$

Ans : (c) The height of a fuzzy set is the highest membership value of the membership function:

$$\text{Height}(A) = \max \mu_A(x_i)$$

A fuzzy set with height 1 is called a normal fuzzy set. In contrast, a fuzzy set whose height is less than 1 is called a subnormal fuzzy set. So, according to the above rule, the fuzzy set A is called normal when $h(A) = 1$

30. An artificial neuron receives n inputs $x_1, x_2, \dots, x_n$ with weights $w_1, w_2, \dots, w_n$ attached to the input links. The weighted sum is computed to be passed on to a non-linear filter ϕ called activation function to release the output.

- (a) $\sum w_i$ (b) $\sum x_i$
(c) $\sum w_i + \sum x_i$ (d) $\sum w_i \cdot x_i$

Ans : (d) Weighted sum is calculated by multiplying inputs with their respective weights and then adding them all. Which can be shown as $\sum w_i \cdot x_i$.

31. Consider the formula in image processing

$$R_D = 1 - \frac{1}{C_R}$$

where $C_R = \frac{n_1}{n_2}$

C_R is called as compression ratio n_1 and n_2 denotes the number of information carrying units in two datasets that represent the same information. In this situation R_D is called as relativeof the first data set.

- (a) Data Compression
(b) Data Redundancy
(c) Data Relation
(d) Data Representation

Ans : (b) Data redundancies is the repetition of data. Data redundancy is a common issue in computer data storage and database system. Data redundancies in database means that some data fields are repeated in the database.

32. Find the false statement :

- (a) In Modern Cryptography, symmetric key algorithms use same key both for Encryption and Decryption.
(b) The symmetric cipher DES (Data Encryption Standard) was widely used in the industry for security product.
(c) The AES (Advanced Encryption Standard) cryptosystem allows variable key lengths of size 56 bits and 124 bits.
(d) Public key algorithms use two different keys for Encryption and Decryption.

Ans : (c) The AES (Advanced Encryption Standard) Cryptosystem allows variable key length of size 128 bits, 192 bits & 256 bits & 56 bit is used DES.

33. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is :
- (a) 110010011001 (b) 11001001
 (c) 110010011001001 (d) 11001001011

Ans : (d) As given polynomial is $x^3 + 1 \Rightarrow 1001$ with this directly we can say option d because we add 3 digits more for the data and this satisfied only by option d.

But general solution could be

$$\begin{array}{r}
 1001 \overline{)11001001000} \quad (1100.011 \\
 \underline{1001} \\
 01011 \\
 \underline{1001} \\
 00100 \\
 \underline{0000} \\
 01000 \\
 \underline{10001} \\
 0001100 \\
 \underline{1001} \\
 01010 \\
 \underline{1001} \\
 0011
 \end{array}$$

34. comparisons are necessary in the worst case to find both the maximum and minimum of n numbers.
- (a) $2n - 2$
 (b) $n + \text{floor } \lg n) - 2$
 (c) $\text{floor} \left(\frac{3n}{2} \right) - 2$
 (d) $2 \lg n - 2$

Ans : (c) $\text{floor} (3n/2) - 2$ comparisons are necessary in the worst case to find both the maximum and minimum of n numbers.

35. Let A and B be two $n \times n$ matrices. The efficient algorithm to multiply the two matrices has the time complexity
- (a) $O(n^3)$ (b) $O(n^{2.81})$
 (c) $O(n^{2.67})$ (d) $O(n^2)$

Ans : (b) The efficient algorithm to multiply the two matrices has the complexity $O(n^{2.81})$ using stressonnr's algorithm.

36. The recurrence relation $T(n) = nT\left(\frac{n}{2}\right) + n^2$ is satisfied by
- (a) $O(n^2)$ (b) $O(n^{\lg n})$
 (c) $O(n^2 \lg n)$ (d) $O(n \lg n)$

Ans: (b) $T(n) = mT(n/2) + n^2$
 now according to master theorem $a = m$, $b = 2$ & $f(n) = \log_2$
 $n \log_2 m > f(n)$ for all $m > 4$ [using case – 1 : master theorem]
 so solution is $O(n \log_2 m)$

37. The longest common subsequence of the sequences $X = \langle A, B, C, B, D, A, B \rangle$ and $Y = \langle B, D, C, A, B, A \rangle$ has length

- (a) 2 (b) 3
 (c) 4 (d) 5

Ans : (c)

S1: ABCBDAB S2: BDCABA LCSRecursive LCSTable

def LCS(S1, S2, x, y):

if $(x = -1)$ or $(y = -1)$

return 0

else if $(S1[x] = S2[y])$

return $1 + \text{LCS}(S1, S2, x-1, y-1)$

else return $\max(\text{LCS}(S1, S2, x-1, y), \text{LCS}(S1, S2, x, y-1))$

Sequence BAAB

		A	B	C	B	D	A	B	
		-1	0	1	2	3	4	5	6
B	-1	0	0	0	0	0	0	0	0
	0	0	0	1	1	1	1	1	1
D	1	0	0	1	1	1	2	2	2
C	2	0	0	1	2	2	2	2	2
A	3	0	1	1	2	2	2	3	3
B	4	0	1	2	2	3	3	3	4
A	5	0	1	2	2	3	3	4	4

38. Assuming there are n keys and each key is in the range $[0, m-1]$. The run time of bucket sort is

- (a) $O(n)$ (b) $O(n \lg n)$
 (c) $O(n \lg m)$ (d) $O(n + m)$

Ans : (d) Assuming there are n keys one each key is in the range $(0, m-1)$. The run time of bucket sort is $O(n)$.

Bucket – sort (A)

1. $n \leftarrow \text{length}[A]$

2. for $i = 1$ to n do

3. insert $A[i]$ into list $B[n \cdot A[i]]$

4. for $i = 0$ to $n-1$ do

5. Sort list B with insertion sort

6. Concatenate the lists $B[0], B[1], B[n-1]$ together in order and their run time is $O(n + m)$

39. A..... complete subgraph and a subset of vertices of a graph $G = (V, E)$ are a clique and a vertex cover respectively.

- (a) minimal, maximal
 (b) minimal, minimal
 (c) maximal, maximal
 (d) maximal, minimal

Ans: (d) Clique– A clique is a subset of vertices of a graph such that every two distinct vertices are adjacent (Every two vertices adjacent means a complete sub graph)

Minimal vertex cover– A minimal vertex cover of a graph is set of minimum vertices such that each edge of a graph is incident to at least one vertex of set.

40. Pumping lemma for context-free languages states: Let L be an infinite context free language. Then there exists some positive integer m such that any $w \in L$ with $|w| \geq m$ can be decomposed as $w = uvxy$ with $|vxy| \leq m$ and $|vy| \geq 1$ such that $UV^zxy^z \in L$ for all $z = 0, 1, 2, \dots$

- (a) $|v| \leq m, |x| \leq 1$ (b) $|v| \leq m, |x| \geq 1$
 (c) $|v| \geq m, |x| \leq 1$ (d) $|v| \geq m, |x| \geq 1$

Ans : (b) Let L be an infinite context free language. There is some positive integer m such that, if S is string of L of length at least m , then

$S = uvwxy$ (for some u, v, w, x, y)

$|vw| \leq m$

$|vx| \geq 1$

$uvwxy \in L$

for all negative of i .

41. The Greibach normal form grammar for the language $L = \{ a^n b^{n+1} \mid n \geq 0 \}$ is :

- (a) $S \rightarrow aSB, B \rightarrow bB \mid \lambda$
 (b) $S \rightarrow aSB, B \rightarrow bB \mid b$
 (c) $S \rightarrow aSB, B \rightarrow b$
 (d) $S \rightarrow aSb \mid b$

Ans : (c) Option (a) & option (b) can't able to generate any string because there is no terminal string present for S . Option (c) can generate all required string ($b, abb, aabbb, \dots$)

42. Given the following statements :

S1: Every context-sensitive language L is recursive.

S2: There exists a recursive language that is not context sensitive.

Which statement is correct?

- (a) $S1$ is not correct and $S2$ is not correct
 (b) $S1$ is not correct and $S2$ is correct
 (c) $S1$ is correct and $S2$ is not correct
 (d) $S1$ is correct and $S2$ is correct

Ans : (d) $S1$ is correct : the set of all turing machines is countable.

$S2$ is also true : $L2 - L1 = L2 \cap L1'$, since $L1$ is recursive $L1'$ is also recursive and hence i.e. also and finally r.e. languages are closed under set difference.

43. What is the bit rate for transmitting uncompressed 800×600 pixel colour frames with 8 bits/pixel at 40 frames/second?

- (a) 2.4 Mbps (b) 15.36 Mbps
(c) 153.6 Mbps (d) 1536 Mbps

Ans : (c) Uncompressed pixel = $800 * 600$
each of 8 bit = $800 * 600 * 8$
Bit rate for transmitting = $800 * 600 * 8 * 40$
= 153600000 = 153.6 Mbps

44. In IPv4, the IP address 200.200.200.200 belongs to

- (a) Class A (b) Class B
(c) Class C (d) Class D

Ans : (c) Consider only the first octet's decimal value. Ignore the rest. The first octet value is 200. Convert the same to binary equivalent. It is 11001000. If the converted bit equivalent has less than 8 bits, then fill its left side with 0's and bring it to a count of 8 bits. The first 3 bits are 110 here and so this IP address belongs to class c address.

45. Which layer of OSI reference model is responsible for decomposition of messages and generation of sequence numbers to ensure correct re-composition from end to end of the network?

- (a) Physical (b) Data-link
(c) Transport (d) Application

Ans : (c) Transport layer is the fourth layer in OSI Reference model. It responds to service requests from session layer & issues service request to network layer. Transport layer is responsible for delivering message between hosts. In transport layer, data travels in the form of segments. Transport layer is responsible for creating an end to end connection between source IP and the destination IP.

46. A client-server system uses a satellite network, with the satellite at a height of 40,000 kms. What is the best-case delay in response to a request? (Note that the speed of light in air is 3,00,000 km/second)?

- (a) 133.33 m sec (b) 266.67 m sec
(c) 400.00 m sec (d) 533.33 m sec

Ans : (d) The request has to go up & down & the response has to go up & come down. The total path length traversed is thus 160,000 km. The speed of light in air and vacuum is 3,00,000 km/sec.
So the propagation delay is $160,000/300,000$ sec = 533.33 m sec.

47. The start and stop bits are used in serial communication for

- (a) error detection
(b) error correction
(c) synchronization
(d) slowing down the communication

Ans : (c) The start and stop bits are used in serial communication for synchronization i.e. to take the receiver data is come after me ready to accept it, and data is finish after me.

48.is a type of transmission impairment in which the signal loses strength due to the resistance of the transmission medium
- (a) Attenuation (b) Distortion
(c) Noise (d) Decibel

Ans : (a) Attenuation is a general term that refers to any reduction in the strength of a signal. Attenuation occurs with any type of signal whether digital or analog. Some times called loss, Attenuation is a natural consequence of signal transmission over long distance.

49. Match the following :

List-I		List-II	
A.	Indexed addressing	1.	is not used when an operand is moved from memory into a register or from a register to memory
B.	Direct addressing	2.	Memory address is computed by adding up two registers plus an (optional) offset
C.	Register addressing	3.	Addressing memory by giving a register plus a content offset
D.	Base-Indexed Addressing	4.	Can only be used to access global variables whose address is known at compile time

- | | | | | |
|-----|----------|----------|----------|----------|
| | A | B | C | D |
| (a) | 2 | 1 | 4 | 3 |
| (b) | 2 | 4 | 1 | 3 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 3 | 1 | 4 | 2 |

Ans : (c)

(a) Indexed addressing is the operand's offset is the sum of the content of an index register S1 or D1 and an 8 bit or 16 bit displacement, so it satisfies (3)

(b) Direct addressing is the operand's offset is given in the instruction as an 8 bit displacement. In this mode the 16 bit effective address of the data is the part of instruction, so it satisfies (4)

(c) Register addressing is placed in one of 8 bit or 16 bit general purpose register. So it satisfies (1)

(d) Base indexed addressing is the operand offset is sum of the content of a base register BX or BP and an index register S1 or D1. (2)

50. Which of the following is a design criteria for instruction formats?
- (a) The size of instructions
(b) The number of bits in the address fields
(c) The sufficient space in the instruction format to express all the operations desired.
(d) All of these

Ans: (d) Design criteria for instruction format (How to atleast the current process or)

1. Size of the instruction balance instruction size vs. Required capabilities, addressing, and inst. fetch times typically word length.
2. Include enough bits to define "All" operation allow for future growth prouder some "unassigned" OP codes.
3. Determine any address range or size limitation of 32-bit address by byte accesses only 4 GB of space the number of RISC register defines the address field size.

51. Synchronization is achieved by a timing device called a which generates a periodic train of

- (a) clock generator, clock pulse
- (b) master generator, clock pulse
- (c) generator, clock
- (d) master clock generator, clock pulse

Ans : (d) Synchronization is achieved by a timing device called a clock pulse generator which generates a periodic train of clock pulses.

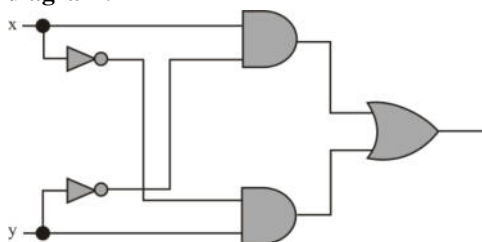
master clock generator also synchronizes all clock

52. Serial access memories are useful in applications where

- (a) Data consists of numbers
- (b) Short access time is required
- (c) Each stored word is processed differently
- (d) None of these

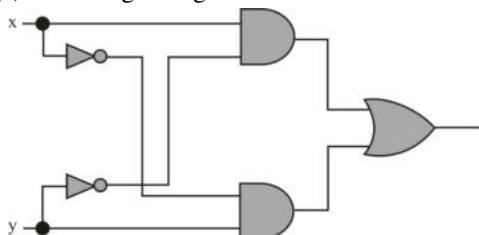
Ans : (d) Serial memories are used where data naturally needs to flow in and out in serial from.

53. What will be the output of the following logic diagram?



- (a) x OR y
- (b) x AND y
- (c) x XOR y
- (d) x XNOR y

Ans : (c) Given logic diagram is



Therefore, function f is:

$$f = (x \cdot \bar{y}) + (\bar{x} \cdot y) = x\bar{y} + \bar{x}y = x \oplus y$$

$$f = x(\text{XOR})y$$

- 54. The essential difference between traps and interrupts is**
- (a) traps are asynchronous and interrupts are synchronous with the program
 - (b) traps are synchronous and interrupts are asynchronous with the program
 - (c) trap are synchronous and interrupts are asynchronous with the I/O devices
 - (d) None of these

Ans : (b) Traps is a software interrupt or synchronous interrupt.
Interrupt is hardware generated or synchronous interrupt
Trap is a software driven and interrupt is hardware originated.

- 55. Consider the following ER diagram:**
- DIAGRam**
- The minimum number of tables required to represent M, N, P, R₁, R₂ is**
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5

Ans : (b) Since R₁ is many to one and participation of M is total, M and R₁ can be combined to form the table {M₁ M₂, M₃, P₁}. N is a week entity set, so it can be combined with P.

- 56. Consider the following schemas:**
- Branch = (Branch-name, Assets, Branch-city)**
- Customer = (Customer-name, Bank name, Customer city)**
- Borrow = (Branch-name, loan number, customer account-number)**
- Deposit = (Branch-name, Account number, Customer-name, Balance)** Using relational Algebra, the Query that finds customers who have balance more than 10,000 is
- (a) $\pi_{customer-name} (\sigma_{balance > 10000} (Deposit))$
 - (b) $\sigma_{customer-name} (\sigma_{balance > 10000} (Deposit))$
 - (c) $\pi_{customer-name} (\sigma_{balance > 10000} (Barrow))$
 - (d) $\sigma_{customer-name} (\pi_{balance > 10000} (Barrow))$

Ans : (a) When use apply projection and use table deposit we get $\pi_{customer-name} (\sigma_{balance > 10000}(Deposit))$

- 57. Find the false statement :**
- (a) The relationship construct known as the weak relationship type was defined by Dey, Storey & Barron (1999).
 - (b) A weak relationship occurs when two relationship types are linked by either Event-Precedent sequence or Condition-Precedent sequence.
 - (c) conceptual model is not accurate representation of "Universe of interest".
 - (d) Ternary, Quaternary and Quintary relationships are shown through a series of application scenario's and vignette's.

Ans: (c) Detailed of requirement to model process using ER modeling grammar and have implicitly assumed that this process takes care of that namely conceptual model. The conceptual model thus developed is a true representation of the "universe of interest" being modeled.

58. Consider the table

Student (stuid, name, course, marks). Which one of the following two queries is correct to find the highest marks student in course 5?

Q1. Select S. stuid

From student S

Where not exist

(Select * from student e where e course = '5' and e marks $\geq$ S marks)

Q2. Select S.stu.id

From student S

Where S.marks > any

(select distinct marks from student S where S.course = 5)

(a) Q1

(b) Q2

(c) Both Q1 and Q2

(d) Neither Q1 nor Q2

Ans : (d) Both the queries are given not correct since, first query will not retrieve any tuple since $\geq$ condition is used. Second query gives stuid of students whose marks are greater than any who is student taking course 5.

Correct queries should be as following:

Q₁: Select S.stuid

From student S

where not exist (select *

From student e

where e. course = '5' and

e.marks > S.marks)

Q₂: Select S.stuid

From student S

where S.marks $\geq$ All (select distinct marks

From student S

where S.course = '5')

59. Armstrong (1974) proposed systematic approach to derive functional dependencies. Match the following w.r.t. Functional dependencies :

List-I		List-II	
A.	Decomposition rule	1.	If $X \rightarrow Y$ and $Z \rightarrow W$ then $\{X, Z\} \rightarrow \{Y, W\}$
B.	Union rule	2.	If $X \rightarrow Y$ and $\{Y, W\} \rightarrow Z$ then $\{X, W\} \rightarrow Z$

C.	Composition rule	3.	If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow \{Y, Z\}$
D.	Pseudo transitivity rule	4.	If $X \rightarrow \{Y, Z\}$ then $X \rightarrow Y$ and $X \rightarrow Z$

	A	B	C	D
(a)	3	2	4	1
(b)	1	3	4	2
(c)	2	1	3	4
(d)	4	3	1	2

Ans : (d)
Decomposition rule is if $x \rightarrow yz$ then $x \rightarrow y, x \rightarrow z$, is the decomposition rule so it satisfies (4)
(b) Union rule is : if $x \rightarrow y, x \rightarrow z$ then $x \rightarrow yz$ so it satisfies (3)
(c) Composition rule is if $x \rightarrow y$ & $z \rightarrow w$ then $xz \rightarrow yw$, so it can satisfies (1)
(d) Pseudo transivity rule can satisfy (2)

60. Match the following :

List-I		List-II	
A.	Secondary Index	1.	Functional Dependency
B.	Nonprocedural Query Language	2.	B-Tree
C.	Closure of set of Attributes	3.	Relational Algebraic Operation
D.	Natural JOIN	4.	Domain Calculus

	A	B	C	D
(a)	1	2	4	3
(b)	2	1	4	3
(c)	1	3	4	2
(d)	2	4	1	3

Ans : (d) It is easy to observe that B-tree has secondary index natural join is relational algebraic operation.
Closure of set of attributes is related to functional dependency.
Non procedural query language is related to domain calculus. hence answer is (d)
Primary index : It is an index on a set of fields that includes the unique primary key for the field and is guaranteed not to duplicates
Secondary index : It is an index that is not a primary index and may have duplicates.

- 61. Which of the following is not true with respect to a trackball and/or spaceball?'**
- I. A trackball is a two dimensional positioning device while as a spaceball provides six degrees of freedom.**
 - II. Unlike the trackball a spaceball does not actually move.**
 - III. A trackball is a three dimensional positioning device while as a spaceball provides six degrees of freedom.**

- (a) I & II (b) II & III
(c) II only (d) III only

Ans : (d) First two statements are correct.

A space ball is a "six degrees of freedom" device. It allows full rotation in multiple directions, such as with a CAD model a trackball, however is merely a standard mouse rather than having the base moved by the mouse platform, you manually actuate the ball. It just moves the cursor around the libsbll page has a nice image showing the spaceball's range and type of motion. Trackball just does xy axis motion

62. Which of the following statement(s) is (are) true?

- I. Two successive translations are additive.**
II. Two successive rotations are additive.
III. Two successive scaling operations are multiplicative.

- (a) I and II (b) I and III
(c) II and III (d) All of the above

Ans : (d) Statement II should be two successive rotations are additive

1. Since translation are just addition by translating vector or displacement hence it is additive
2. Since rotation are done with some angles. So two successive rotations are additive (angles will be added or subtracted)
3. Scaling is done by multiplying by scaling vector so every scaling will be multiplicative

63. Given below are three basic rules :

- (I) Squash and Stretch**
(II) Slow-in and Slow-out
(III) To stage the action properly

These rules are applied in case of

- (a) Rendering (b) Morphing
(c) Animation (d) All of these

Ans : (c) The animation were made with little or no reference to nature out of these classes grace a new way of drawing moving human figures and animals.

The principle and rules of animation is :

- (i) Timing
- (ii) ARCS
- (iii) Anticipation
- (iv) Squash & stretch
- (v) Appeal
- (vi) Slow in slow out
- (vii) To stage the action properly

64. Which of the following points lies on the same side as the origin, with reference to the line $3x + 7y = 2$?

- (a) (3, 0) (b) (1, 0)
(c) (0.5, 0.5) (d) (0.5, 0)

Ans : (d) The X and Y intercept of line. $3X + 7Y = 2$ on axes are $(2/3, 0)$, $(0, 2/7)$ so all points greater than $(0.66, 0.28)$ will lie on the other side of line with respect to

origin and all which are less than (0.66, 0.28) will lie on the same side as origin.

So only point 0.5 & 0 will be lie on the same side as origin

65. The transformation matrix required for conversion of CMY colour model to RGB colour model is given as:

$$(a) \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} C \\ M \\ Y \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$(b) \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} C \\ M \\ Y \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$(c) \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$

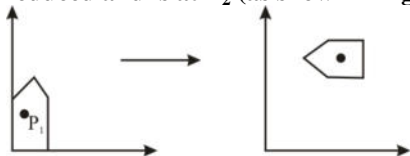
$$(d) \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} C \\ M \\ Y \end{bmatrix} - \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Ans : (c) The RGB model is an additive model; ie. the colour present in the light add to form new colours.

The CMY(Cyan-Magenta-Yellow) model is a subtractive model appropriate to absorption of colours.

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

66. What steps shall be required to rotate an object about the point P_1 (as shown in fig. 1) and its placement such that what was at P_1 is now reduced and is at P_2 (as shown in fig. 2) ?



I. Translate P_1 to origin

II. Scale as required

III. Rotate

IV. Translate to the final position P_2 .

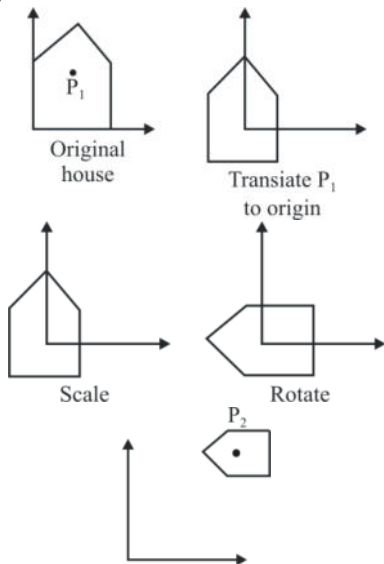
(a) I, II and III

(b) II, III and IV

(c) I, III and IV

(d) All of the above

Ans : (d)



Rotation of the house about point P_1 , and placement such that what was at P_1 is at P_2 .

67. In Unix, how do you check that two given strings a and b are equal?

- (a) test \$a – eq \$b
- (b) test \$a – equal \$b
- (c) test \$a = \$b
- (d) Sh – C test \$ a = = \$ b

Ans : (c) In unix, test \$ a = \$b returns true if the string are same.

68. In windows 2000 operating system all the processor-dependent code is isolated in a dynamic link library called

- (a) NTFS file system
- (b) Hardware abstraction layer
- (c) Microkernel
- (d) Process Manager

Ans : (b) Hardware abstraction layer hardware abstraction layer is a logical division of code that serves as an abstraction layer between computer physical hardware and its software. It provides a driven interface allowing a program to communicate with the hardware. The main purpose of hardware abstraction layer is to conceal different hardware architectures from the OS by providing a uniform interface to the system peripherals.

69. To place a sound into a word document, following feature of windows is used :

- (a) Clip board
- (b) Task switching
- (c) C Win App
- (d) OLE

Ans : (d)

- Object linking and embedding (OLE) allows embedding & linking to document & other objects.
- The main objecting of OLE is to add different kinds of data to a document from different applications.
- This creates a compound document and a master file to which the document makes reference.

70. Translation Look-aside Buffer (TLB) is:

- (a) a cache-memory in which item to be scratched is compared one-by-one with the keys.
- (b) a cache-memory in which item to be searched is compared with all the keys simultaneously.
- (c) an associative memory in which item to be searched is compared one-by-one with the keys.
- (d) an associative memory in which item to be searched is compared with all the keys simultaneously.

Ans : (d) Translation look aside buffer (TLB) is an associative memory in which item is to be searched is compared with all the keys simultaneously. When the associative memory is presented with an item, the item is compared with all keys simultaneously. If the item is found, the corresponding value field is returned.

71. Simplest way of deadlock recovery is :

- (a) Roll back
- (b) Preempt resource
- (c) Lock one of the processes
- (d) Kill one of the processes

Ans : (d) Kill one of the processes recovery through killing processes is the crudest but simplest way to break a deadlock.

Kill one of the processes in the deadlock cycle : the other processes get its resources.

Choose process that can be return from the beginning.

72. The directory structure used in Unix file system is called

- (a) Hierarchical directory
- (b) Tree structured directory
- (c) Directed acyclic graph
- (d) Graph structured directory

Ans : (c) Unix file system directory structure is a DAG due to presence of hard links which makes sharing of files & sub directories among different directories possible

73. Which statement is not true about process O in the Unix operating system?

- (a) Process O is called init process.
- (b) Process O is not created by fork system call.
- (c) After forking process1, process O becomes swapper process.
- (d) Process O is a special process created when system boots.

Ans : (a) Process O is called init process answer is (a)

74. Which of the following commands would return process _id of sleep command?

- (a) Sleep 1 and echo \$?
- (b) Sleep 1 and echo \$#
- (c) Sleep 1 and echo \$x
- (d) Sleep 1 and echo \$!

Ans : (d) Sleep 1 & echo \$!

\$\$	PID of current shell
\$!	PID of last background command
\$?	exit status of last command
\$1	Name of current's command first parameter
\$9	Name of current command's Ninth parameter
\$@	All of current command's parameters
\$*	All of current command's parameters can't preserving white space & quoting

So answer is (d)

75. Possible thread states in Windows 2000 operating system include:

- (a) Ready, running and waiting
- (b) Ready, standby, running, waiting, transition and terminated
- (c) Ready, running, waiting, transition and terminated
- (d) Standby, running, transition and terminated

Ans : (b) A thread in **Windows 2000** can be in one of the six possible states, namely, ready, standby, running, waiting, transition and terminated. The standby state and transition state are somewhat unique to Windows 2000. Other states of a thread (i.e., ready, running, waiting and terminated) are quite common across all the operating systems.